

ZNOTES.ORG

UPDATED TO 2023-25 SYLLABUS

CAIE IGCSE MATHS

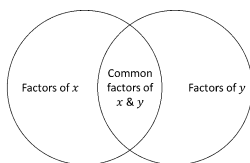
SUMMARIZED NOTES ON THE THEORY SYLLABUS

1. Number

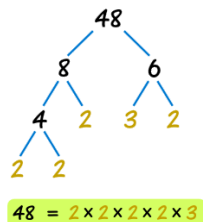
- Natural numbers:**
 - used for counting purposes
 - all possible rational & irrational numbers
- Integer:** a whole number
- Prime numbers:**
 - divisible only by itself and one
 - 1 is not a prime number
- Rational numbers:** can be written as a fraction
- Irrational numbers:** cannot be written as a fraction e.g. π
- Cube numbers:** made from multiplying a rational number to itself twice.
- Reciprocals:** A number made by raising a rational number to -1, or 1 over that number

1.2. HCF and LCM

- Highest Common Factor and Lowest Common Multiple:**



- HCF = product of common factors of x and y
- LCM = product of all items in Venn diagram

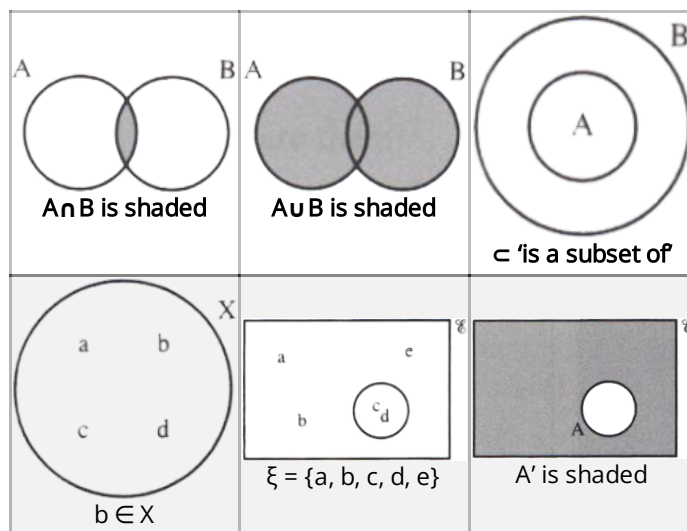
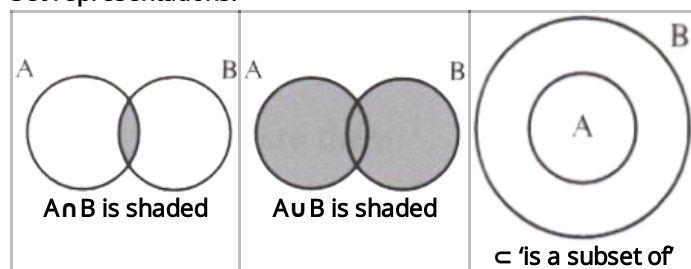


- Prime Factorization:** finding which prime numbers
 - multiply together to make the original number

1.3. Sets

- Definition of sets e.g.
 - $A = \{x: x \text{ is a natural number}\}$
 - $B = \{(x, y): y = mx + c\}$
 - $C = \{x: a \leq x \leq b\}$
 - $D = \{a, b, c, \dots\}$

Set representations:



of elements in A

- $\in$ = ...is an element of...
- $\notin$ = ...is not an element of...
- A' = complement of set A
- $\emptyset$ or $\{\}$ = empty set
- ξ = Universal set
- $A \cup B$ = union of A and B
- $A \cap B$ = intersection of A and B
- $A \subseteq B$ = A is a subset of B
- $A \subset B$ = A is a proper subset of B
- $A \not\subseteq B$ = A is not a subset of B

1.4. Indices

Standard form:

- $10^4 = 10000$
- $10^3 = 1000$
- $10^2 = 100$
- $10^1 = 10$
- $10^0 = 1$
- $10^{-1} = 0.1$
- $10^{-2} = 0.01$
- $10^{-3} = 0.001$
- $10^{-4} = 0.0001$
- $10^{-5} = 0.00001$

Limits of accuracy:

- The degree of rounding of a number
 - E.g. 2.1 to 1 d.p $2.05 \leq x < 2.15$
- Finding limits when adding/multiplying: add/multiply respective limits of values
- Finding maximum value possible when dividing/subtracting: max value divided by/minus min value
- Finding minimum value possible when dividing/subtracting: min value divided by/minus max value

1.5. Ratio & Proportion

- **Ratio:** used to describe a fraction
 - e.g. 3 : 1
- **Foreign exchange:** money changed from one currency to another using proportion
 - E.g. Convert \$22.50 to Dinars
 $\$1 : 0.30\text{KD}$
 $\$22.50 : 6.75\text{KD}$
- **Map scales:** using proportion to work out map scales
 - 1km = 1000m
 - 1m = 100cm
 - 1cm = 10mm
- **Direct variation:** y is proportional to x

$$y \propto x$$

$$y = kx$$

- **Inverse variation:** y is inversely proportional to x

$$y \propto \frac{1}{x}$$

$$y = \frac{k}{x}$$

1.6. Percentages

- **Percentage:**
 - Convenient way of expressing fractions
 - Percent means per 100
- **Percentage increase or decrease:**

$$\text{Percentage increase} = \frac{\text{Actual Change}}{\text{Original Amount}} \times 100$$

- **Simple interest:**

$$I = \frac{PRT}{100}$$

Where, $P = \text{Principal}$, $R = \text{Rate Of Interest}$, and $T = \text{Time}$

- **Compound interest:**

$$A = P \left(1 + \frac{R}{100} \right)^n$$

Where, $P = \text{Principal}$, $R = \text{Rate Of Interest}$, and $T = \text{Time}$

1.7. Speed, Distance & Time

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

- **Units of speed:** km/hr or m/s
- **Units of distance:** km or m
- **Units of time:** hr or sec

$$\text{km/hr} \times \frac{5}{18} = \text{m/sec}$$

$$\text{m/sec} \times \frac{18}{5} = \text{km/hr}$$

2. Algebra & Graphs

2.1. Factorisation

- **Common factors:**

$$3x^2 + 6x$$

$$3x(x + 2)$$

- **Difference of two squares:**

$$25 - x^2$$

$$(5 + x)(5 - x)$$

- **Group factorization:**

$$4d + ac + ad + 4c$$

$$4(d + c) + a(c + d)$$

$$(4 + a)(c + d)$$

- **Trinomial:**

$$x^2 + 14x + 24$$

$$x^2 + 12x + 2x + 24$$

$$x(x + 12) + 2(x + 12)$$

$$(x + 2)(x + 12)$$

2.2. Quadratic Factorization

- **General equation:**

$$ax^2 + bx + c = 0$$

- **Solve quadratics by:**

- Trinomial factorization
- Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- When question says, "give your answer to two decimal places", **use formula!**
- Derivation of the Quadratic Formula is the same as saying "Make x the subject in $ax^2 + bx + c = 0$ "

$$ax^2 + bx + c = 0$$

Factorize a out

$$a \left(x^2 + \frac{b}{a}x \right) + c = 0$$

Complete the Square

$$a \left(\left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a^2} \right) + c = 0$$

$$a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c = 0$$

$$a \left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a}$$

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$x + \frac{b}{2a} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x + \frac{b}{2a} = \frac{\pm \sqrt{b^2 - 4ac}}{\sqrt{4a^2}}$$

Note: $4a^2$ is a square number

$$x + \frac{b}{2a} = \frac{\pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

• **Standardized form:**

- $y = ax^2 + bx + c$

• **Complete Square form:**

- $y = (x + a)^2 + b$ (Where axis of symmetry is $x = -a$)

- To find turning point of quadratic equation, complete the square, then the turning point is: $(-a, b)$

• **Ways to solve Quadratic equation:**

- Graphing Method
- Factorizing
- Quadratic Formula
- Complete the Square

- **Graphing Method** – Graph the equation, see where it touches the x-axis

- **Factorizing**

e.g. $x^2 - x - 6 = 0$

$$x^2 - x - 6 = 0$$

$$(x - 3)(x + 2) = 0$$

$$x_1 = 3$$

$$x_2 = -2$$

- **Quadratic Formula**

e.g. $x^2 - x - 6 = 0$

Where $a = 1, b = -1, c = -6$

Plug the numbers in the Quadratic Formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Therefore:

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)}$$

$$x_1 = 3$$

$$x_2 = -2$$

- **Complete the Square**

e.g. $x^2 + 10x + 5 = 0$

(WARNING! Coefficient of x^2 Must be 1 for this to work)

$$x^2 + 10x + 5 = 0$$

$$(x + 5)^2 - 5^2 + 5 = 0$$

$$(x + 5)^2 - 20 = 0$$

$$(x + 5)^2 = 20$$

$$x + 5 = \pm \sqrt{20}$$

$$x = -5 \pm \sqrt{20}$$

Answer is:

$$x_1 = -5 + \sqrt{20}, x_2 = -5 - \sqrt{20}$$

2.3. Reciprocal Graphs (Hyperbola)

• **Standardized Form:**

- $y = \frac{a}{x}$

If a is Positive : The Line will be in the 1st&3rd Quadrant	If a is Negative : The Line will be in the 2nd&4th Quadrant
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2.4. Cubic Equation

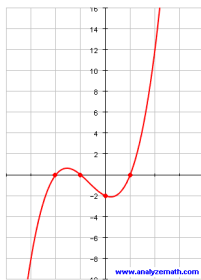
• **Standardized Form:**

- $y = ax^3 + bx^2 + cx + d$

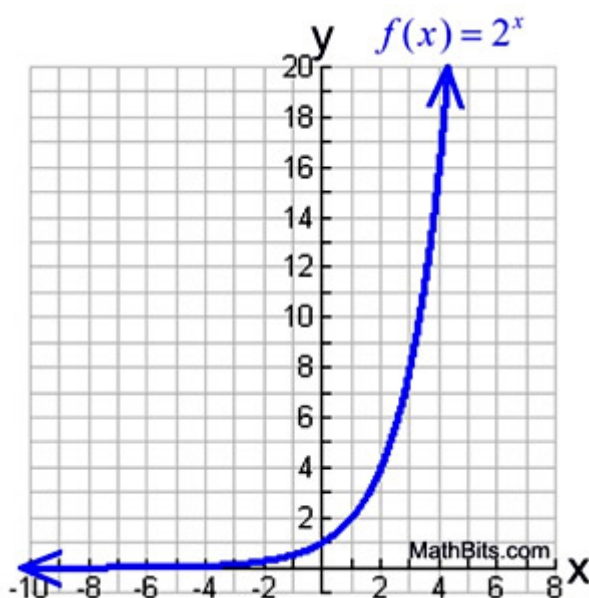
• **Properties:**

- Highest Exponent of x is 3
- Has a maximum of 2 turning points

Turning points are points after which a graph changes its gradient's sign, therefore changing direction between up or down



2.5. Exponential Graphs



- Standardized form:

- $y = a(b)^x$

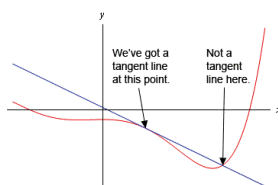
- Properties:

- a is the y -intercept
 - Asymptotes are lines that a curve approaches, but never touches because the curve continues to infinity. In this case, The asymptotes are $y = 0$ and $x = 4$
 - b is the rate of growth
 - When $0 < b < 1$, the graph will go downwards from left to right

2.6. Gradient of a Curve

- By drawing tangents

- In a straight line, gradient is constant
 - Curves have varying gradients throughout the graph. To find the gradient at a point:
 - Draw the graph
 - Draw a tangent at the point in the graph, ensuring it only touches the graph at that point (Use a ruler)
 - Find the gradient of the tangent



- Using differentiation

- $\frac{dy}{dx}$ gives you the gradient of the curve at any point in terms of x
 - When $y = x^n$, $\frac{dy}{dx} = nx^{n-1}$
 - Stationary/ turning point: $\frac{dy}{dx} = 0$
 - 1st Derivative = $\frac{dy}{dx} = f'(x)$
 - 2nd Derivative = $\frac{d^2y}{dx^2} = f''(x)$
 - To determine if stationary point is maximum or minimum:
 - Use 2nd derivative
 - Maximum point: $\frac{d^2y}{dx^2} < 0$
 - Minimum point: $\frac{d^2y}{dx^2} > 0$
 - Use gradients around the point
 - Input x values slightly above and below stationary point and calculate gradient

2.7. Simultaneous Equations

- Can be solved either by substitution or elimination
- Generally solved by substitution as follows:
 - Step 1: obtain an equation in one unknown and solve this equation
 - Step 2: substitute the results from step 1 into linear equation to find the other unknown
- The points of intersection of two graphs are given by the solution of their simultaneous equations

2.8. Inequalities

- Solve like equations
- Multiplying or dividing by negative $\Rightarrow$ switch sign

$$\frac{y}{-3} \geq -7$$

$$y \leq -7 \times -3$$

$$y \leq 21$$

- When two inequalities present, split into two

$$x < 3x - 1 < 2x + 7$$

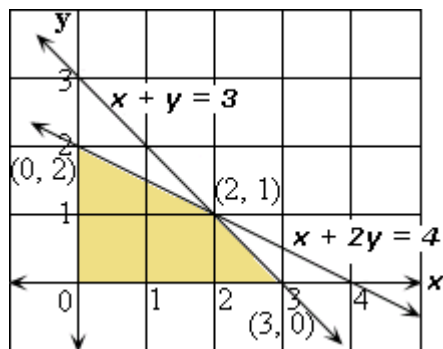
$x < 3x - 1$	$3x - 1 < 2x + 7$
$x > \frac{1}{2}$	$x < 8$

$$\frac{1}{2} < x < 8$$

2.9. Linear Programming

- For strict inequalities ($<$, $>$) use broken line
- For non-strict inequalities ($\leq$, $\geq$) use solid line
- Steps to solve:
 - Interpret $y = mx + c$
 - Draw straight line graphs
 - Shade

- Solve



2.10. Sequences

- Linear sequences:** Find common difference e.g. 3, then multiply by n and work out what needs to be added
- Quadratic sequences:**
 - Format: $an^2 + bn + c$

$$\begin{array}{rcl} a + b + c & = & 2 \\ 3a + b & = & 6 \\ 2a & = & 4 \end{array}$$

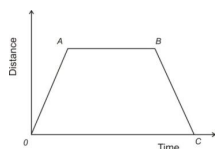
- Work out the values and then place into formula to work out n th term formula

- Geometric progression:** sequence where term has been multiplied by a constant to form next term

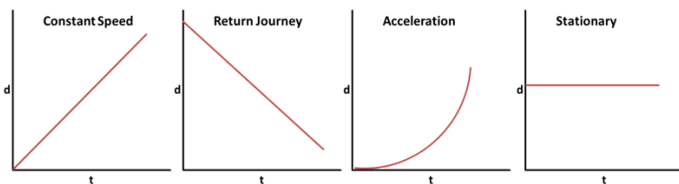
$$n\text{th term of G.P.} = ar^{(n-1)}$$

- $a = 1^{\text{st}}$ term $r = \text{common ratio}$

2.11. Distance-Time Graphs

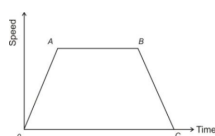


- From O to A:** Uniform speed
- From B to C:** Uniform speed (return journey)
- From A to B:** Stationary (speed = 0)

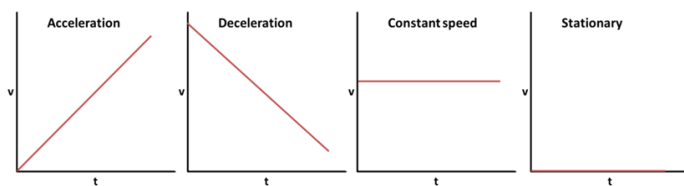


- $\text{Gradient} = \text{speed}$

2.12. Speed-Time Graphs



- From O to A:** Uniform increase in speed
- From A to B:** Constant speed (acceleration = 0)
- From B to C:** Uniform deceleration / retardation



- Area under a graph = distance travelled.
- Gradient = acceleration.
- If the acceleration is negative, it is called deceleration or retardation. (moving body is slowing down.)

2.13. Functions

- Function notation:**
 - $f : x \rightarrow 2x - 1$
 - Function f such that x maps onto $2x - 1$
- Composite function:** Given two functions $f(x)$ and $g(x)$, the composite function of f and g is the function which maps x onto $f(g(x))$
- $f(2)$
 - Substitute $x = 2$ and solve for $f(x)$
- $fg(x)$
 - Substitute $x = g(x)$
- $f^{-1}(x)$
 - Let $y = f(x)$ and make x the subject

3. Geometry

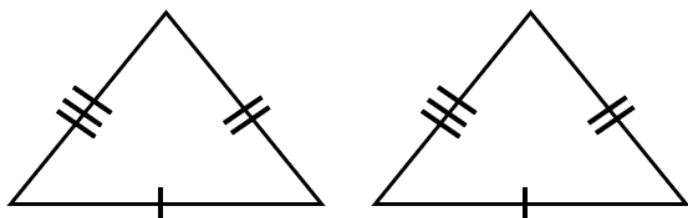
3.1. Similarity

- Similarity can be worked out by the AAA (Angle - Angle - Angle) rule.
- AAA (Angle - Angle - Angle) rule:** All the corresponding angles of the triangles must be equal.

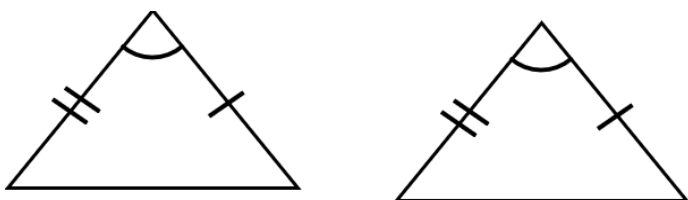


3.2. Congruence

- SSS (Side - Side - Side) rule:** All the three sides of the triangles must be equal



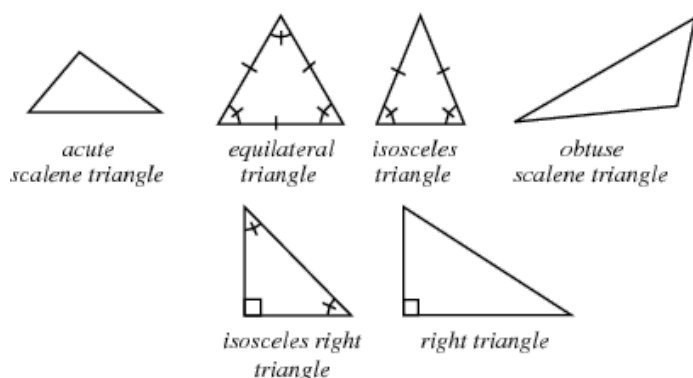
- **RHS (Right angle – Hypotenuse – Side) rule :**
- There must be two right-angled triangles
- The length of the hypotenuses must be the same
- One of the corresponding sides of each triangle must be the same
- **SAS (Side – Angle – Side) rule:**
- There must be an angle and a side present
- The angle of the adjacent sides must be equal
- The two sides of the triangle must be equal



- **ASA (Angle – Side – Angle) rule:** The sides adjacent to the equal angles must be of the same length.

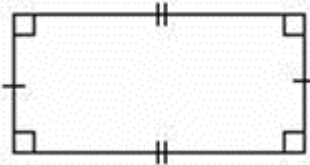
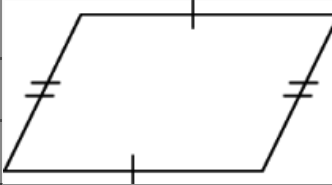
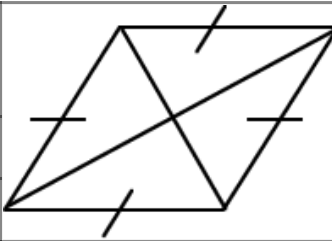
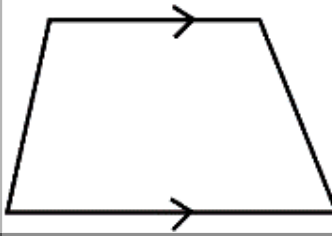
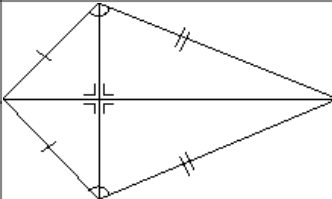


3.3. Triangles



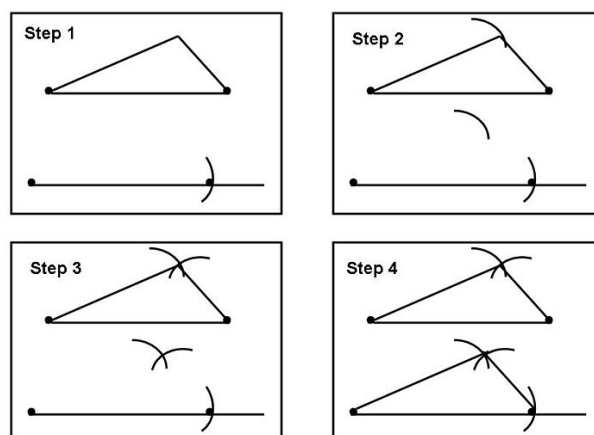
3.4. Quadrilaterals

Rectangle:	
Opposite sides parallel/equal	
all angles 90°	

Rectangle:	
diagonals bisect each other	
Parallelogram:	
Opposite sides parallel/equal	
opposite angles equal	
diagonals bisect each other	
Rhombus:	
A parallelogram with all sides equal	
opposite angles equal	
diagonals bisect each other	
Trapezium:	
One pair of sides parallel	
Kite:	
Two pairs of adjacent sides equal	
diagonals are perpendicular to each other	

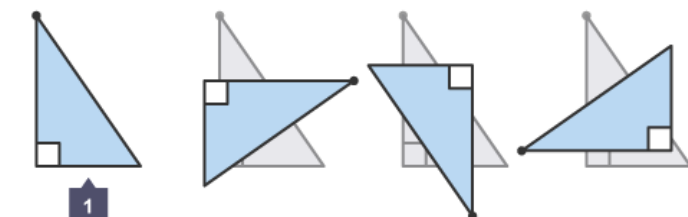
3.5. Construction

- **Constructing triangles:**



3.6. Symmetry

- **Line of symmetry:** Divides a two-dimensional shape into two congruent (identical) shapes
- **Plane of symmetry:** Divides a three-dimensional shape into two congruent solid shapes
- The number of times shape fits its outline during a complete revolution is called the order of **rotational symmetry**

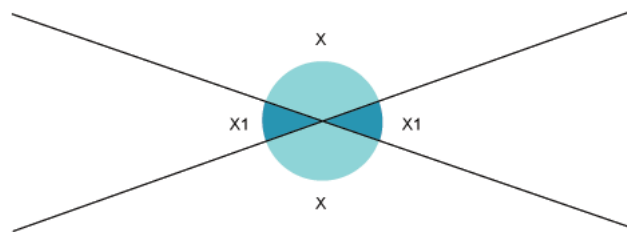


Shape	Number of Lines of Symmetry	Rotational Symmetry Order
Square	4	4
Rectangle	2	2
Parallelogram	0	2
Rhombus	2	2
Trapezium	0	1
Kite	1	1
Equilateral triangle	3	3
Regular hexagon	6	6

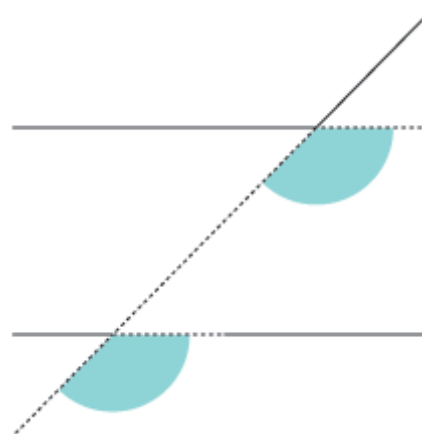
- **Properties of circles:**
 - Equal chords are equidistant from the centre
 - The perpendicular bisector of a chord passes through the centre
 - Tangents from an external point are equal in length

3.7. Polygons

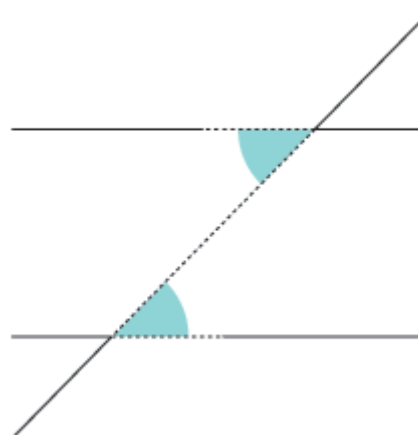
- Sum of angles at a point = 360°
- Angles on a straight line = 180°
- Sum of angles in a triangle = 180°
- For regular polygon
 - External angles = $\frac{360^\circ}{n}$
 - Internal angles = $180^\circ - \frac{360^\circ}{n}$
- For irregular polygon:
 - Sum of exterior angles = 360°
 - Sum of interior angles = $180(n - 2)$
- Vertically opposite angles are equal



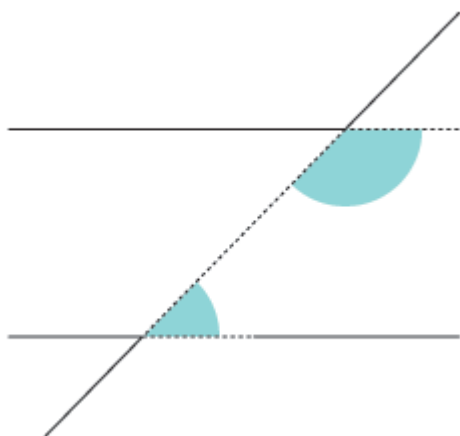
- Corresponding angles are equal



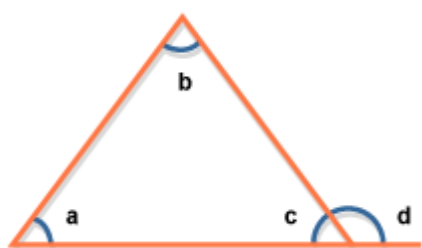
- Alternate angles



- Co-interior angles add up to 180°



- Exterior angle = sum of interior opposite $\angle$



3.8. Circle Theorem

Angle at center = twice angle on circumference	Angle subtended by same arc at circumference are equal
Angles in semicircle are 90°	Opposite angles in a cyclic quadrilateral = 180°
Tangents from one point are equal. $\angle$ between tangent and radius is 90°	Alternate segment theorem

4. Mensuration

4.1. Area

- Parallelogram = $b \times h$ or $AB \sin \theta$
- Triangle = $\frac{1}{2}b \times h$

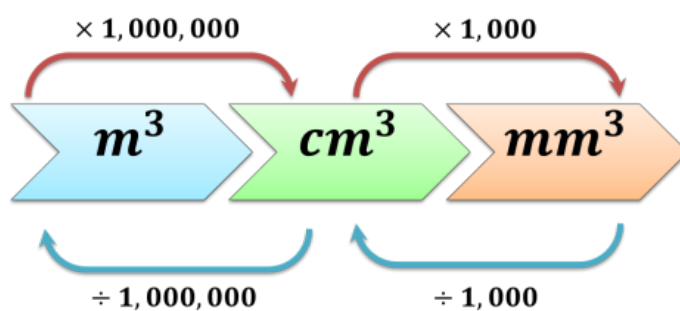
- Trapezium = $\frac{1}{2}(a + b)h$
- Circle = πr^2
- Sector = $\pi r^2 \times \frac{\theta}{360}$

4.2. Volume and Surface Area

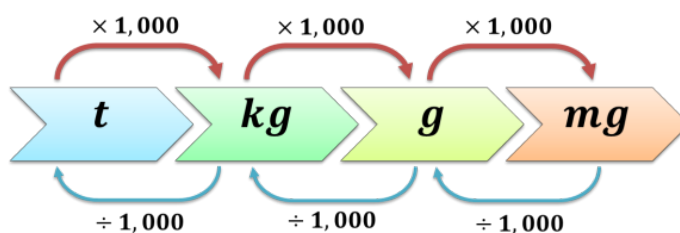
- Cuboid
 - Surface Area = $2(lw + hw + hl)$
 - Volume = hlw
- Cylinder
 - Curved surface area = $2\pi rh$
 - Volume = $\pi r^2 h$
- Cone
 - Curved surface area = πrl
 - Volume = $\frac{1}{3}(\pi r^2 h)$
- Sphere
 - Surface Area = $4\pi r^2$
 - Volume = $\frac{4}{3}\pi r^3$
- Hemisphere
 - Surface area = $3\pi r^2$
 - Volume = $\frac{2}{3}\pi r^3$
- Kite
 - Surface area = $\frac{pq}{2}$

4.3. Units

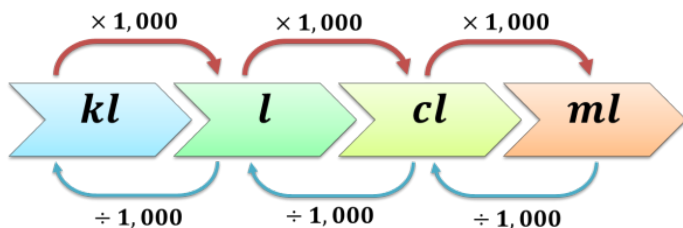
- Volume:



- Mass:



- Capacity:



- Connecting volume and capacity:

- $1ml = 1cm^3$
- $1kl = 1m^3$

Density = $\frac{\text{Mass}}{\text{Volume}}$

5. Coordinate Geometry

5.1. Graphs

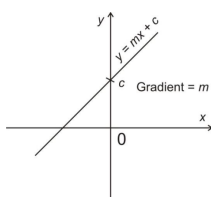
- Gradient of a Straight Line:

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

- Equation of Line:

$$y = mx + c$$

- Find the gradient, m
- Find the y -intercept, c



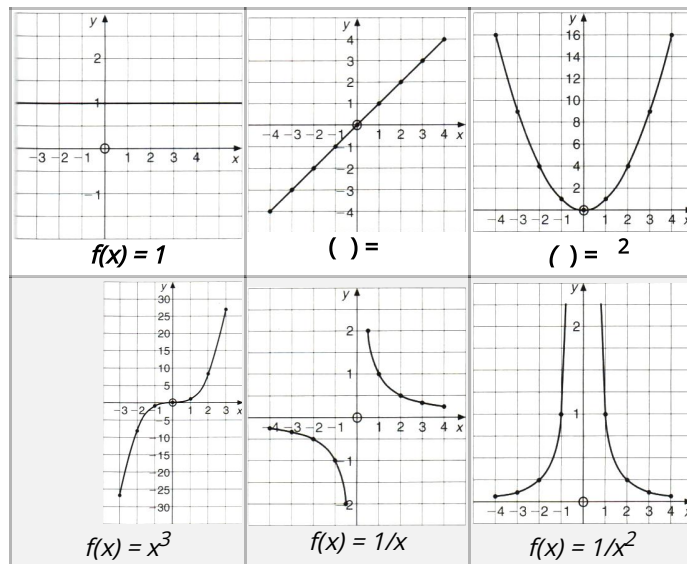
- Midpoint of Graph:

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- Length between two points:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

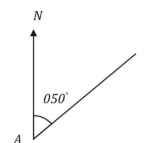
5.2. Sketching Graphs



6. Trigonometry

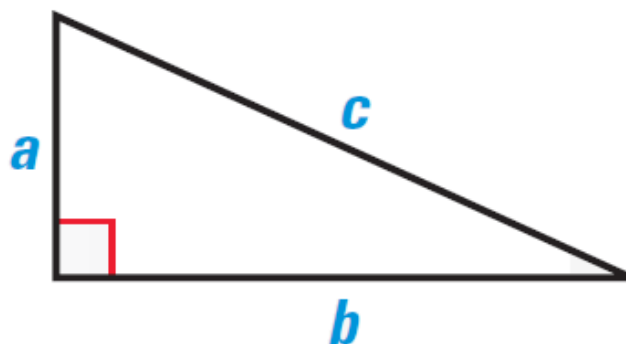
6.1. Bearings

- The bearing of a point B from another point A is:
 - An angle measured from the north at A.
 - In a clockwise direction.
 - Written as three-figure number (i.e. from 000° to 360°)
- e.g. The bearing of B from A is 050°

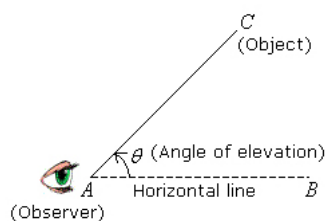


6.2. Pythagoras Theorem

- To find hypotenuse
 - $a^2 + b^2 = c^2$

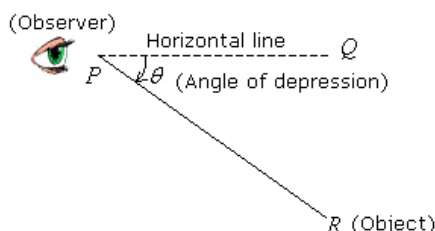


- To find one of the shorter sides
 - $a^2 = c^2 - b^2$
 - $b^2 = c^2 - a^2$
- Angle of elevation:
 - Angle above the horizontal line



• **Angle of depression:**

- Angle below the horizontal line.

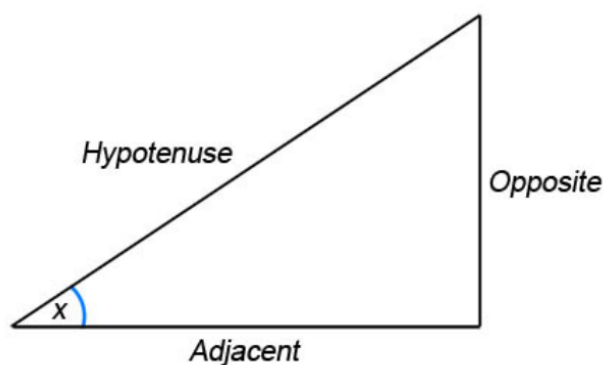


- Area of a triangle: $\frac{1}{2}ab \sin c$

6.3. Ratios

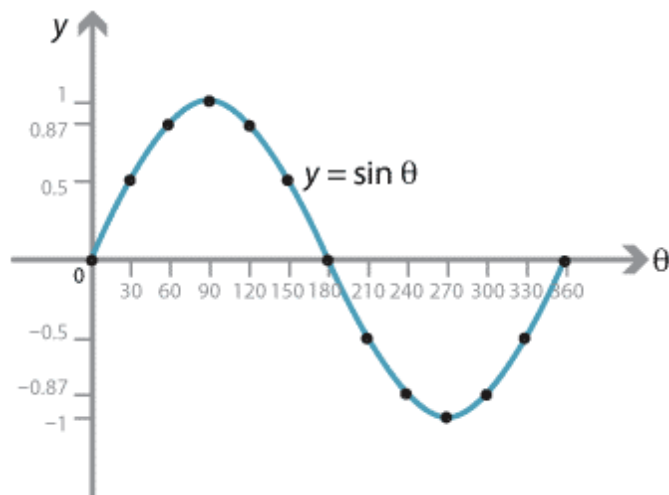
Right angled triangles:

- $\sin x = \frac{\text{opposite}}{\text{hypotenuse}} \rightarrow \text{SOH}$
- $\cos x = \frac{\text{adjacent}}{\text{hypotenuse}} \rightarrow \text{CAH}$
- $\tan x = \frac{\text{opposite}}{\text{adjacent}} \rightarrow \text{TOA}$

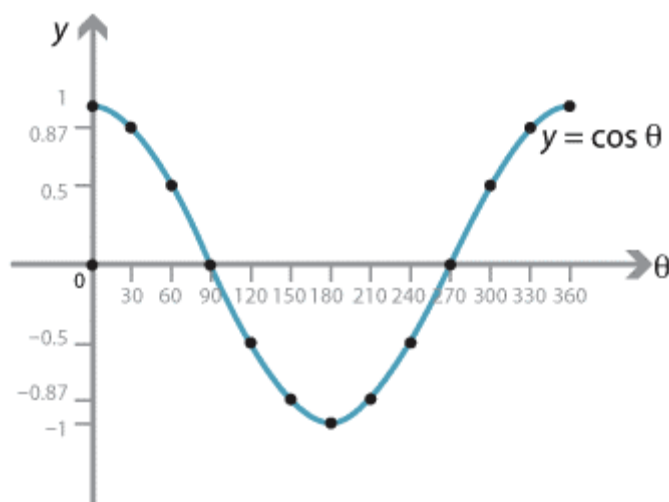


6.4. Graphs of simple trigonometric functions

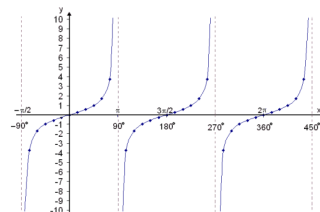
- $\sin(x) = \sin(180^\circ - x)$



- $\cos(x) = \cos(360^\circ - x)$



- Sine and cosine shifted by 90°
- Sine has x-intercepts at multiples of 180° , and cosine at $(90^\circ + \text{multiples of } 180^\circ)$
- $\tan(x) = \tan(180^\circ + x)$



- Goes to infinity at $90^\circ, 270^\circ, 450^\circ, \dots$
- Has x-intercepts at multiples of 180°

6.5. Sine & Cosine Rules

- Sine rule:

$$\frac{A}{\sin a} = \frac{B}{\sin b} = \frac{C}{\sin c}$$

- Cosine rule

- To find the angle given 3 sides

$$\cos a = \frac{b^2 + c^2 - a^2}{2bc}$$

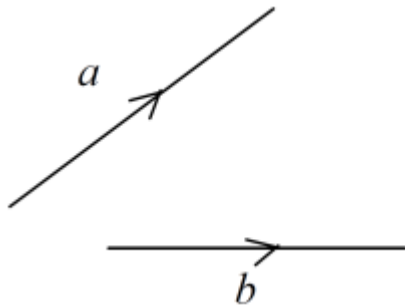
- To find side given angle and two sides

$$a^2 = b^2 + c^2 - 2bc \cos a$$

7. Vectors & Transformations

7.1. Vectors

- Vector quantity** has both magnitude and direction
 - E.g. Vectors a and b represented by the line segments, can be added using 'parallelogram rule' or 'nose-to-tail method'



- Multiplication by a scalar:**
 - Scalar quantity:** has a magnitude but no direction
 - The negative sign reverses the direction of the vector
- Column vector:**

$$\begin{pmatrix} x \\ y \end{pmatrix}$$

- Top number = horizontal component
- Bottom number = vertical component
- Parallel vectors:**
 - Vectors are parallel if they have the same direction
 - In general, the vector $k \begin{pmatrix} a \\ b \end{pmatrix}$ is parallel to $\begin{pmatrix} a \\ b \end{pmatrix}$
- Modulus of a vector:**
 - In general, if $x = \begin{pmatrix} m \\ n \end{pmatrix}$, $|x| = \sqrt{m^2 + n^2}$

7.2. Transformation

- Reflection (M):**
 - When describing a reflection, the position of the mirror line is essential
- Rotation (R):**
 - The centre, angle and direction of rotation are needed to describe a rotation
 - A clockwise rotation is negative, and an anticlockwise rotation is positive

- Translation (T):**

$$\begin{pmatrix} x \\ y \end{pmatrix}$$

- When describing a translation, it is necessary to give the translation vector
- Enlargement (E):**
 - To describe an enlargement, state the scale factor, K and the centre of enlargement

$$\text{Scale factor} = \frac{\text{length of image}}{\text{length of object}}$$

$$\text{Area of image} = K^2 \times \text{area of object}$$

- If $K > 0$, both object and image lie on same side of the centre of enlargement
- If $K < 0$, object and image lie on opposite side of the centre of enlargement

8. Probability

- Probability is the study of chance, or the likelihood of an event happening
-

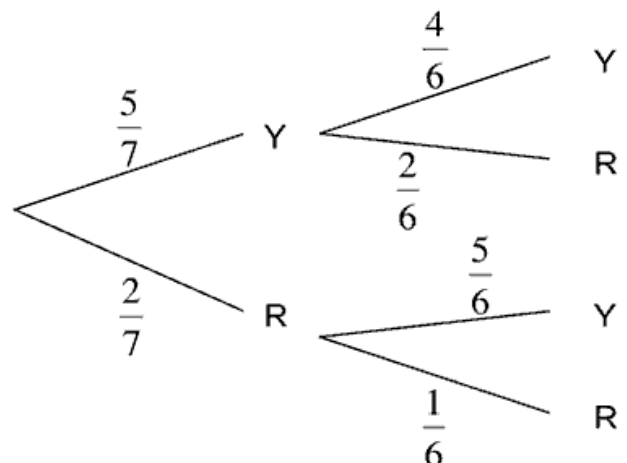
$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

- If probability = 0, event is impossible
- If probability = 1, event is certain to happen
- All probabilities lie between 0 and 1

8.2. Events

Exclusive events:

- Two events are exclusive if they cannot occur at the same time



- The OR Rule:**

- For exclusive events A and B
- $P(A \text{ or } B) = P(A) + P(B)$

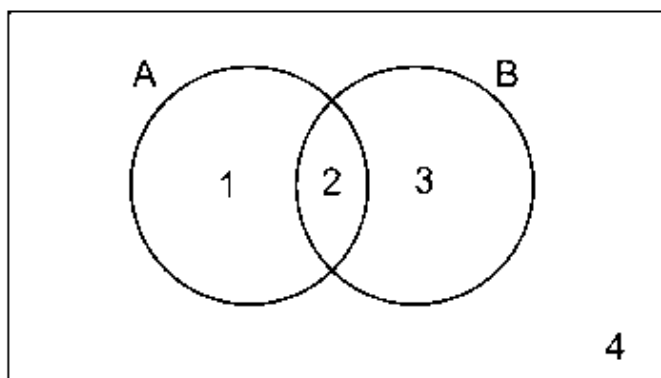
Independent events:

- Two events are independent if occurrence of one is unaffected by occurrence of other
- **The AND Rule:**
 - $P(A \text{ and } B) = P(A) \times P(B)$

8.3. Conditional Probability

- Probability of an event (A), given that another (B) has already occurred

Symbol : $P(A|B)$

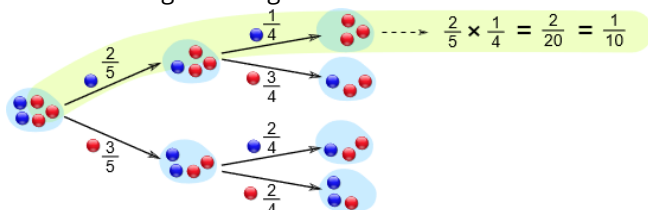


$P(A|B)$ is A given B

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2}{2+3} = \frac{2}{5}$$

- Calculate using Venn diagram:
- Construct the Venn diagram, using sample space of both events
 - $P(A|B) = P(A \cap B) / P(B)$

- Calculate using tree diagrams:

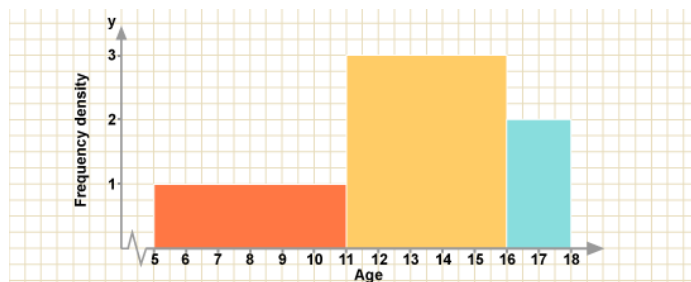


- Construct tree diagram.
- Write the outcomes of the first event
- Connect both the second and first events outcome
- Write probability on top of each event's line
- Multiply probabilities on the lines to the required outcome
- Note: The probabilities reduce with each step if objects are replaced
- Calculate using two-way tables:
 - Column and row headers are the sample space of the two events
 - Fill in each cell with the correct number of outcomes

- Take the required number from the table and divide by the sum of all values in the row/column of the condition provided.
- Remember: $P(A|B)$ and $P(B|A)$ are not the same

9. Statistics

9.1. Histograms



- **Histogram:** Displays frequency of continuous or grouped discrete data in the form of bars
- Bars are joined together and may be of varying width
- Frequency of the data is represented by the area of the bar and not the height
- When class intervals are different, area of the bar represents the frequency, not the height
- Frequency density plotted on y-axis, not frequency
- **Class width** = Interval
- **Frequency density** = Height

$$\text{Frequency} = \text{Class width} \times \text{Frequency density}$$

9.2. Averages

- **Mean**

$$\frac{\text{Sum of values}}{\text{number of values}}$$

- **Median:**

- The middle value - when the data has been written in ascending or descending order
- Odd no. of values $\frac{5+1}{2} = 3\text{rd value}$
- Even no. of values $\frac{6+1}{2} = 3.5\text{th value}$ (add two values divide by 2)

- **Mode:**

- Most frequently occurring value

- **Range:**

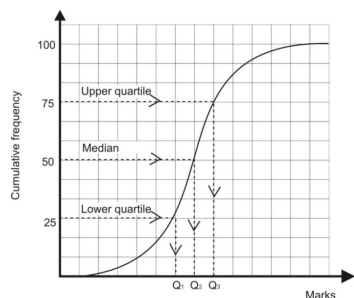
- Difference between highest and lowest values

- **Estimated mean of grouped data:**

- Work out midpoints of each group and multiply by frequency
- Divide by number of values

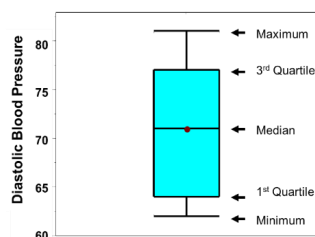
9.3. Cumulative Frequency

- Cumulative frequency is the total frequency up to a given point
- Inter-quartile range = *upper quartile* – *lower quartile*

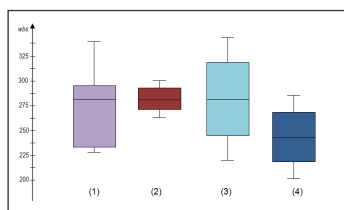


9.4. Box-and-whisker plots

- Construction
 - Find median and two quartiles
 - Draw three lines of equal width along these values
 - Complete the boxes
 - Draw 'whiskers' extending from the box to the maximum and minimum values.
 - Draw two more lines at the ends



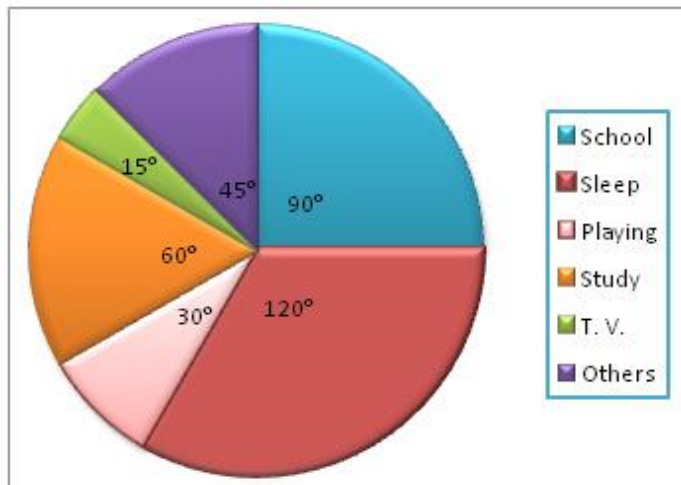
- Interpretation:
 - Median, quartiles and extreme values can be found by reading on the scale of y-axis
 - Short boxes mean low IQR and vice versa (2), (3)
 - Long whiskers mean a lot of extreme values and vice versa (1)
 - Difference in position of boxes represents if data in one set is overall higher or lower than another data set. (3) and (4)
 - Variation in lengths of different sections and position of median show how evenly the data is spread, compared to other data sets (1)



9.5. Pie Charts

- Sectors represent data, and these sectors form a circle.
- Angle of a sector:

$$\theta = \frac{\text{Number of an item}}{\text{Total number of items}} \times 360^\circ$$



- Sum of angles in a pie chart is 360°

9.6. Stem and Leaf diagrams

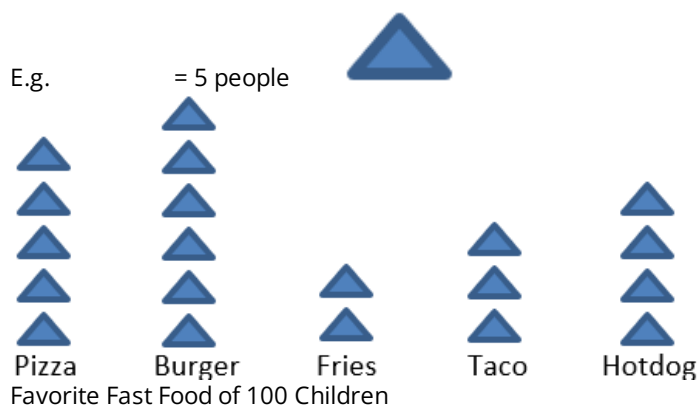
- Stem-and-Leaf diagram is a quick way of summarizing a range of data.
- There is a column known as the stem, contains which contains unique elements of data formed by removing last digits of the data.
- Keys are used in this diagram

stem	leaf
0	1, 1, 2, 2, 3, 4, 4, 4, 4, 5, 8
1	0, 0, 0, 1, 1, 3, 7, 9
2	5, 5, 7, 7, 8, 8, 9, 9
3	0, 1, 1, 1, 2, 2, 2, 4, 5
4	0, 4, 8, 9
5	2, 6, 7, 7, 8
6	3, 6

Key: 6|3 = 63 years old

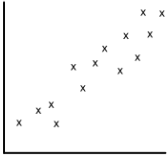
9.7. Pictograms

- Data is represented in pictures
- A key is given to represent the value of a picture.

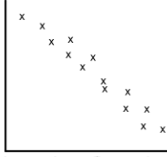


9.8. Scatter Diagrams

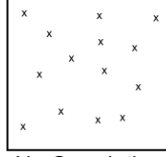
- Displays the correlation between two sets of data
- May have positive, negative or no correlation



Positive Correlation

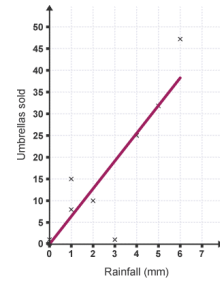


Negative Correlation



No Correlation

- Line of best fit drawn through points that has an equal number of points on each side to show the trend



CAIE IGCSE

Maths

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